

Survival Analysis for Topological Data Analysis Features

Asim Dey

University of Texas at El Paso

joint with

Yulia R. Gel, UT Dallas

and

Snigdhanu Chatterjee, University of Minnesota

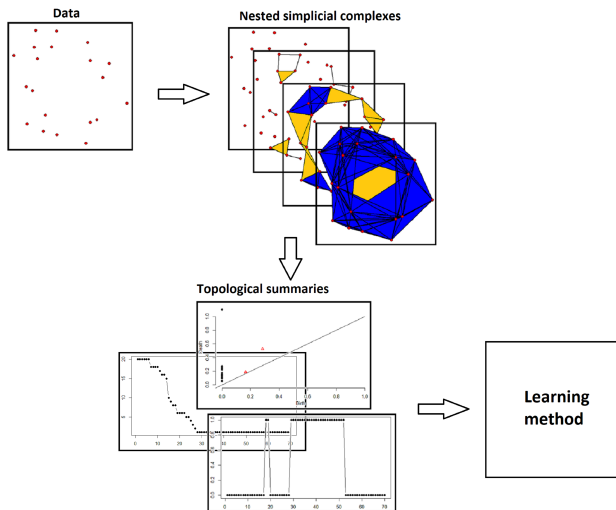
- 1 Intro to Topological Data Analysis (TDA)
- 2 Survival Analysis of TDA features
- 3 Application – Comparing Multivariate Distributions
- 4 Size and Power comparison of the TDA based Tests

Topological data analysis (TDA)

- TDA is the emerging methodology at the interface of algebraic topology, machine learning, statistics, and data science.
- TDA enhances our understanding on the **hidden role of geometry and topology** in **organization and functionality** of a complex system.
- Three key advantages:
 - Topology studies shapes in a *coordinate-free* manner
 - Topology studies properties of shapes that are invariant to *minor perturbations*
 - Topology deals with *compressed representations of shapes*

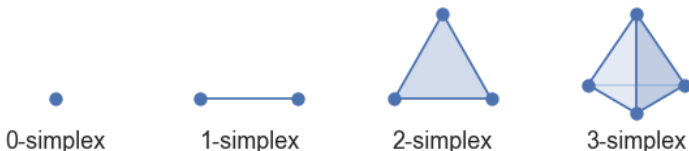
The ultimate goal is to explore intrinsic data structure via systematic mathematical machinery that offers a robust description of shapes.

Standard TDA pipeline in applications



Topological data analysis (TDA)

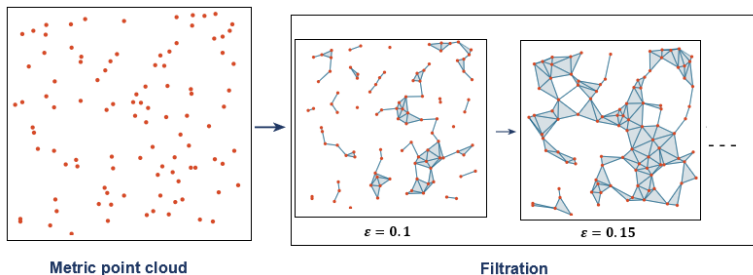
Let X be a point cloud in a metric space $(\mathbb{X}, \mathbb{D})$. An **abstract simplicial complex** over X is a collection $\mathcal{C}$ of finite subsets of X such that if $\sigma \in \mathcal{C}$, then $\tau \in \mathcal{C}$ for all $\tau \subseteq \sigma$. If $|\sigma| = p + 1$, then σ is called a p -simplex.



- Simplicial complex is formed by “gluing” many simplices together such that the intersection of any two simplices is also a simplex.
- Commonly used simplicial complexes: Čech, Vietoris-Rips.

Vietoris-Rips (VR) Complex/Filtration

Vietoris-Rips (VR) complex: $VR_\epsilon = \{\sigma \subset X \mid \mathbb{D}(x_i, x_j) \leq \epsilon, \forall x_i, x_j \in \sigma\}$.



- An increasing sequence of scales $0 < \epsilon_1 < \epsilon_2 < \dots < \epsilon_n$ results in a hierarchically nested sequence of graphs $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \dots \subseteq \mathcal{G}_K$.
- For each $\mathcal{G}_k$, $k = 1, \dots, K$, we identify topological features.

Signal vs Noise

Q: Which extracted topological features are **persistent** over the filtration? And which ones are just a topological noise?

The idea is that features persisting over varying thresholds ϵ_j are considered as **topological signals**. Features with a short lifespan may be viewed as **topological noise**.

The three main topological summaries are:

- Betti numbers
- Barcode plots
- Persistent Diagrams (PD)

Betti numbers

- **Homology** is an algebraic topological invariant that, roughly speaking, describes the holes in a space.
- The p -dimensional holes (connected components, loops, trapped volumes, etc.) of a topological space X are encoded in an algebraic structure called the p -th **homology group** of X , denoted $H_p(X)$.
- The rank of this group is referred to as the p -th **Betti number**, β_p , and counts the number of independent p -dimensional holes.
- The Betti- p number of a **simplicial complex** $\mathcal{C}(\epsilon)$ of dimension d corresponding to a filtration level ϵ , denoted by $\beta_p(\mathcal{C}(\epsilon))$, is defined as the rank of the p -th homology group of $\mathcal{C}(\epsilon)$, $H_p(\mathcal{C}(\epsilon))$, $p = 0, 1, 2, \dots, d$.

Betti numbers

- Informally, the *Betti numbers* represent counts of connected components and p -dimensional holes.
- β_0 = number of connected components
- β_p = number of p -dimensional holes

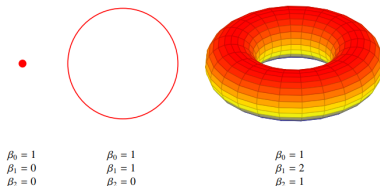


Figure: Betti number for a point (left), circle (center), and torus (right).

Barcode plots and Persistence Diagrams (PD)

- Given a nested sequence of topological spaces $X_1 \subseteq X_2 \subseteq \dots \subseteq X_n$, the inclusion $X_i \subseteq X_{i'}$ for $i \leq i'$ induces a linear map $H_p(X_i) \rightarrow H_p(X_{i'})$ on the corresponding p -th homology for all $p \geq 0$.
- The idea of **persistent homology** is to track elements of $H_p(X_i)$ as the scale parameter i (ϵ) increases.
- A standard way to represent persistent homology information is a **Barcode** plot or **persistence diagram** (PD).
- We say that a topological feature is born at i in the **filtration process** if $\mathcal{C}(\epsilon_i^b)$ is the first complex where we record it, and we say that a topological feature dies at j if we last observe this feature in $\mathcal{C}(\epsilon_j^d)$.

Barcode plots and Persistence Diagrams (PD)

- Barcodes depict the birth and death times of each topological feature as a bar.
- The persistence diagram summarizes the birth and death times of topological features through a multi-set $\mathcal{D} = \{(\epsilon_i^b, \epsilon_j^d) \in \mathbb{R}^2 | \epsilon_i^b < \epsilon_j^d\}$.

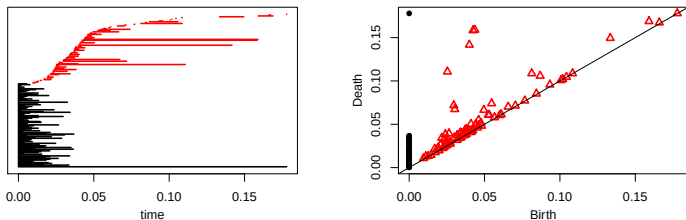


Figure: Barcodes plot (left panel) and Persistent Diagram ($\mathcal{D}$) right panel for a 100-point cloud.

- Black lines in the barcode plot represent the lifetime of connected components and the red lines represent the lifetime of loops.
- Black points in $\mathcal{D}$ represent connected components and red triangles represent loops.

Lifetime distribution

- Consider a persistence diagram $\mathcal{D}$ with N points:

$$\left\{ \{(\epsilon_i^b, \epsilon_j^d) \in \mathbb{R}^2 \mid \epsilon_i^b < \epsilon_j^d\} \right\}_{l=1}^{l=N}.$$

- We can define the survival time of each feature in the filtration process as

$$S_l = \epsilon_{li}^b - \epsilon_{lj}^d,$$

where $l = 1, 2, \dots, N$.

- Let T be a nonnegative random variable representing the lifetimes of individual TDA features defined over the interval $[0, \epsilon_n]$.
 - Let $f(t)$ denote the probability density function (pdf) of T .
 - The cumulative distribution function (cdf) of t can be defined as

$$F(t) = P_r(T \leq t)$$

Lifetime distribution

- The probability of an individual feature surviving to time t is given by the survival function:

$$S(t) = P_r(T \geq t). \quad (1)$$

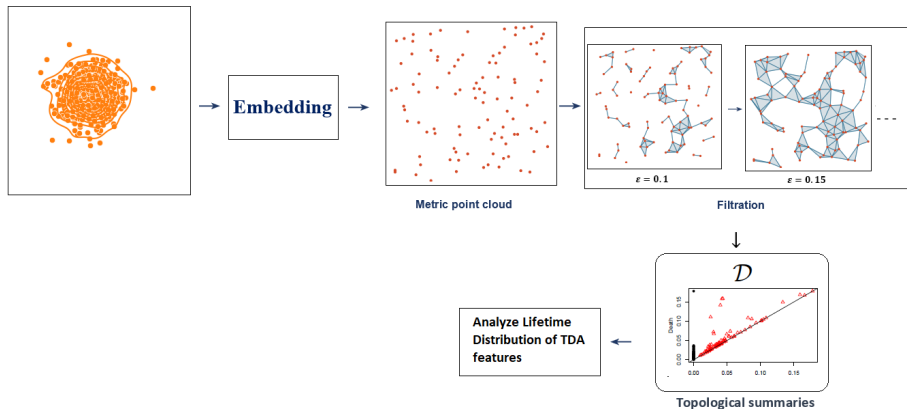
- Hazard function $h(t)$, i.e., the instantaneous rate of death or failure at time t is defined as

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P_r(t \leq T \leq t + \Delta t | T \geq t)}{\Delta t} \quad (2)$$

- Cumulative hazard function $H(t)$ is defined as

$$H(t) = \begin{cases} \int_0^t h(x) dx & \text{for continuous case} \\ \sum_{j: t_j < t} h(t_j) & \text{for discrete case} \end{cases} \quad (3)$$

Illustration of the Survival Analysis of TDA features



Comparing Multivariate Distributions

- For univariate cases, two popular tests are:
 - 1 Kolmogorov-Smirnov (K-S) test, and
 - 2 Anderson-Darling test.

Note: Multivariate tests for comparing two distributions are not very common. One option could be Hotelling T2 (but it is a multivariate mean test).

Objectives:

- The goal is to apply the lifetime distribution of TDA features for comparing multivariate distributions.
- In particular, we want to develop a test based on the lifetime distribution of TDA features to compare high dimensional multivariate distributions.

Experiment 1: Two 2D Multivariate Gaussian Models

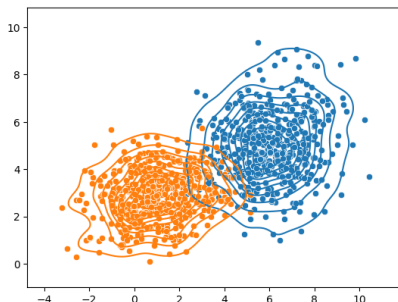
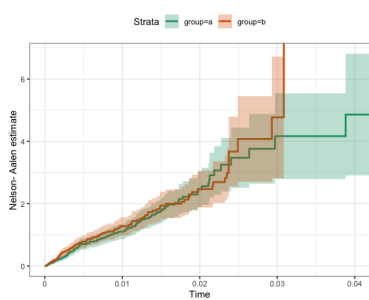
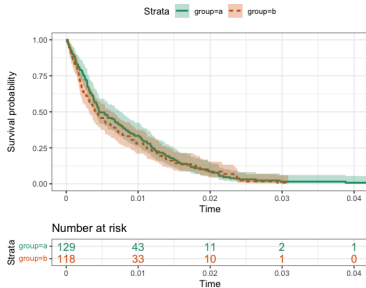
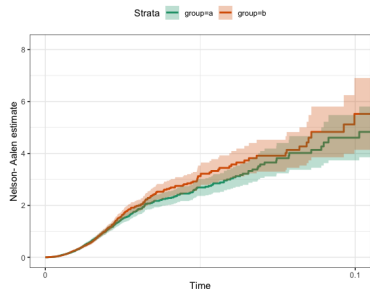
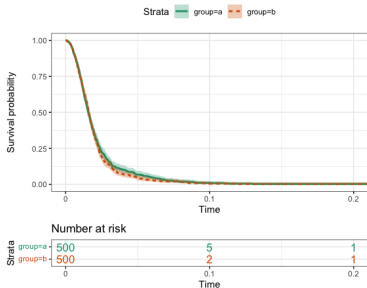


Figure: Scatter plot with contour lines for 3 separate 2D Multivariate Gaussian Model – Case 2.

Steps:

- 1 We embed the Multivariate Gaussian data sets into metric spaces.
- 2 We apply TDA filtration and obtain persistence diagrams $\mathcal{D}$.
- 3 From $\mathcal{D}$ we compute the lifetimes of each TDA features.
- 4 Using the obtained lifeline observations we estimate different survival analysis functions, e.g., $S(t)$, $F(t)$, and $H(t)$.

Experiment 1: Two 2D Multivariate Gaussian Models



Experiment 2: Three 2D Multivariate Gaussian Model

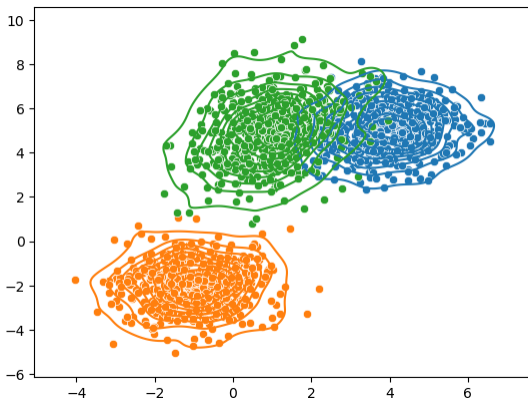
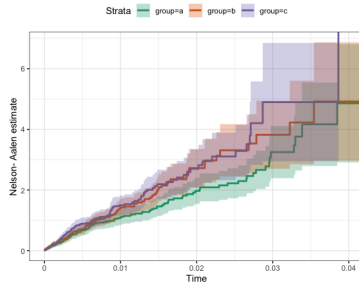
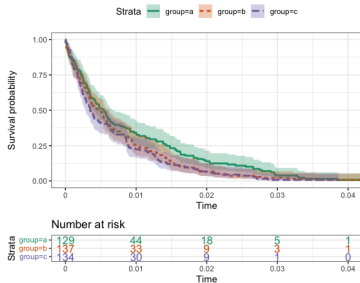
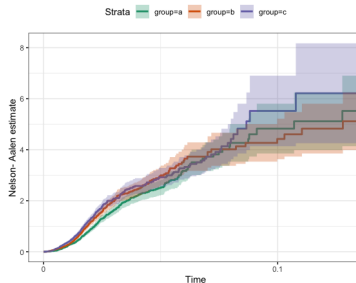
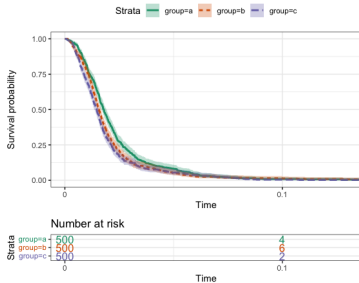


Figure: Scatter plot with contour lines for 3 separate 2D Multivariate Gaussian Model.

Experiment 2: Three 2D Multivariate Gaussian Model



Size and Power of the TDA based Test

We develop three tests based on the lifetime distributions of TDA features:

- 1 TDA Log-Rank test
- 2 TDA Kolmogorov-Smirnov
- 3 TDA Anderson–Darling test

Size and Power of the TDA based Tests – Two multivariate distributions:

- $\mathbf{X}_1 \sim N_p(\mu_1, \Sigma_1)$,
- $\mathbf{X}_2 \sim N_p(\mu_2, \Sigma_2)$,
- Sample sizes are $n_1 = n_2 = 100$,
- $\alpha = 0.05$.

Each case is simulated 1000 times to approximate the power and size of the respective test.

Size and Power of the TDA based Test

	Probabilities of rejecting H_0		
	TDA based Log-Rank test	TDA based Anderson-Darling test	TDA based Kolmogorov-Smirnov
p=5			
Size of the test			
$\mu_1 = \mu_2$, and $\Sigma_1 = \Sigma_2$	0.11	0.05	0.01
Power of test:			
$\mu_1 \neq \mu_2$, but $\Sigma_1 = \Sigma_2$	0.15	0.09	0.00
$\mu_1 = \mu_2$, and $\Sigma_1 \neq \Sigma_2$	0.24	0.35	0.15
$\mu_1 \neq \mu_2$, and $\Sigma_1 \neq \Sigma_2$	0.22	0.37	0.16
p=10			
Size of the test			
$\mu_1 = \mu_2$, and $\Sigma_1 = \Sigma_2$	0.16	0.13	0.03
Power of test:			
$\mu_1 \neq \mu_2$, but $\Sigma_1 = \Sigma_2$	0.22	0.19	0.06
$\mu_1 = \mu_2$, but $\Sigma_1 \neq \Sigma_2$	0.4	0.63	0.45
$\mu_1 \neq \mu_2$, and $\Sigma_1 \neq \Sigma_2$	0.3	0.61	0.36

THANK YOU!